*Queen's University*
*48 University Avenue*
*Kingston, Canada K7L 3N8*
*d.miyamoto@queensu.ca*
*ian-david-miyamoto.github.io*
*Citizenship: Canadian*

# David Miyamoto

## Current position

2025– **Coleman Postdoctoral Fellow**, *Queen's University*, Kingston, Canada
Working group: Andrew Lewis

## Employment

2023–2025 **Postdoctoral Researcher**, *Max Planck Institute for Mathematics*, Bonn, Germany
Working group: Christian Blohmann

## Education

2018–2023 **PhD Mathematics**, *University of Toronto*, Toronto, Canada
Supervisor: Yael Karshon
Thesis: Geometry of leaf spaces of singular foliations

2017–2018 **MSc Mathematics**, *University of Toronto*, Toronto, Canada
Supervisor: Dmitry Faifman
Thesis: Characterizing $U(1,1)$ and translation-invariant generalized convex valuations on $\mathbb{C}^2$

2013–2017 **BSc Mathematics (Minor in History)**, *University of Toronto*, Toronto, Canada

## Publications and Preprints

### Published

2026 *Lie groupoids determined by their orbit spaces*
David Miyamoto
*J. Noncommut. Geom.* (2026) 687, 37 pages

2025 *Quasifold groupoids and diffeological quasifolds*
Yael Karshon and David Miyamoto
*Transform. Groups* 30 (2025), no. 3, 1333-1367

2024 *Riemannian foliations and quasifolds*
Yi Lin and David Miyamoto
*Math. Z.* 308 (2024), no. 49, 32 pages

– *Diffeological submanifolds and their friends*
Yael Karshon, David Miyamoto, and Jordan Watts
*Differential Geom. Appl.* 96 (2024), 14 pages

– *Singular foliations through diffeology*
David Miyamoto
*Recent advances in diffeologies and their applications*, 139–160, Contemporary Mathematics vol. 794, published by the American Mathematical Society (2024). MR4712602

2023 *The basic de Rham complex of a singular foliation*
David Miyamoto
Int. Math. Res. Not. IMRN (2023), no. 8, 6364–6401. MR4574377

### Submitted

2026 *Differentiation of tepui groupoids and the holonomy groupoid of singular foliations*
Alfonso Garmendia, David Miyamoto, and Leonid Ryvkin
Preprint, 32 pages. arXiv:2611.08825

– *Diffeological Riemannian orbifolds*
David Miyamoto
Preprint, 27 pages. arXiv:2607.08939

2025 *A singular Serre-Swan theorem via tepui fibrations*
Alfonso Garmendia, David Miyamoto, and Leonid Ryvkin
Preprint, 33 pages. arXiv:2510.20936

– *Lie algebras of quotient groups*
David Miyamoto
Preprint, 44 pages. arXiv:2502.10260. To appear in *J. Lie Theory*

### In Preparation

*The tangent category of elastic diffeological spaces*
Christian Blohmann and David Miyamoto

## Research Presentations

### Conference talks

2026 Aug. "Diffeological Riemannian orbifolds" (poster)
Summer School Poisson 2026
Universiteit Antwerpen, Antwerp, Belgium

2026 Mar. "Lie's third theorem via elastic diffeological groups (two talks)"
Building up Differential Homotopy Theory 2026
Shinshu University, Matsumoto, Japan

2025 Jan. "Lie algebras of diffeomorphism groups of non-compact manifolds (part 2)"
Infinite-dimensional Geometry: Theory and Applications
Erwin Schrödinger International Institute for Mathematics and Physics, Vienna, Austria

2024 Oct. "A diffeological approach to integrating Lie algebras"
Exploring new arrows in the BGW groupoid
Retreat in Bielefeld, Germany

2024 July "Lie groupoids determined by their orbit spaces"
Workshop on Hamiltonian Geometry and Quantization
Fields Institute, Toronto, Canada

2024 Mar. "Lie groupoids determined by their orbit spaces"
Higher Geometric Structures Along the Lower Rhine XVII
Max Planck Institute for Mathematics, Bonn, Germany

2024 Mar. "Lie groupoids determined by their orbit spaces"
Building-up Differential Homotopy Theory
Osaka Metropolitan University, Osaka, Japan

2023 Nov. "Leaf spaces of Killing foliations"
Atelier on Higher Structures in Differential Geometry (poster)
Institut Camille Jordan, Lyon, France

2023 Jan. "The basic complex of a singular foliation"
Workshop on Lie Groups, Singular Spaces, and Higher Structures
Fields Institute, Toronto, Canada

2022 July "Quasifolds as groupoids and as diffeological spaces"
Poisson 2022 (poster)
Instituto de Ciencias Mathematicas, Madrid, Spain

2022 July "Quasifold groupoids and diffeological quasifolds"
AMS-EMS-SMF Joint International Meeting, Diffeology session
Université Grenoble Alpes, Grenoble, France

2022 Apr. "Basic forms on foliated manifolds"
Gone Fishing 2022
Georgia Southern University, Savannah, United States

2019 Dec. "Basic forms on foliated manifolds"
CMS Winter Meeting (poster)
Toronto, Canada

## Seminar talks

In reverse chronological order

2026
- Topology and Geometry Seminar
  IIIT Delhi, Delhi, India
- Special seminar (Job talk)
  Georgia Southern University, Savannah, Georgia

2025
- Algebra and Geometry Seminar
  Queen's University, Kingston, Canada
- Geometry Seminar
  Central Michigan University, Mount Pleasant, United States
- Departmental Colloquium
  Central Michigan University, Mount Pleasant, United States
- Introduction to Diffeology / Introduction to Higher Structures in Geometry Seminar
  Queen's University, Kingston, Canada
  – A series of talks
- Dynamics, Geometry and Groups Seminar
  Queen's University, Kingston, Canada
- Oberseminar Geometrie, Topologie und Analysis
  Universität zu Köln, Köln, Germany
- Sino-Russian Mathematics Center-JLU Colloquium
  Jilin University, Jilin, China
  – Three talks
- Global Diffeology Seminar
  Online
- Geometry Seminar
  KU Leuven, Leuven, Belgium

2024
- Higher Differential Geometry Seminar
  Max Planck Institute for Mathematics, Bonn Germany
- Oberseminar Deformationsquantisierung und Geometrie
  Universität Würzburg
- Special Seminar
  Université de Lorainne

2023
- Global Diffeology Seminar
  Online
- Symplectic Geometry Seminar
  University of Toronto, Toronto, Canada
- MPI–Oberseminar
  Max Planck Institute for Mathematics, Bonn, Germany
- MPIM Topology Seminar
  Max Planck Institute for Mathematics, Bonn, Germany
- Higher Differential Geometry Seminar
  Max Planck Institute for Mathematics, Bonn Germany
- Symplectic and Poisson Geometry Seminar
  University of Illinois Urbana-Champaign, Urbana-Champaign, United States

2022
- Graduate Student Seminar
  University of Toronto, Toronto, Canada
- Symplectic Geometry Seminar
  University of Toronto, Toronto, Canada
- Geometry and Dynamics Seminar
  Tel Aviv University, Tel Aviv, Israel
- Symplectic and Poisson Geometry Seminar
  University of Illinois Urbana-Champaign, Urbana-Champaign, United States
- Local Poisson Mini-Conference
  University of Toronto, Toronto, Canada

2021
- Global Diffeology Seminar
  Online
- Graduate Student Seminar
  University of Toronto, Toronto, Canada

2020
- Graduate Student Seminar
  University of Toronto, Toronto, Canada

## Awards and Grants

2025 Fields Sponsored Postdoctoral Fellowship at Queen's University

2023 Queen Elizabeth II/Lloyd George Elliott Graduate Scholarship in Science and Technology
For academic excellence

2022 Ida Bulat Memorial Graduate Fellowship
For academic excellence

2021 Daniel B. DeLury Teaching Assistant Award
For excellence in teaching

2019 Coxeter Graduate Scholarship
For academic excellence

2019 AARMS Award
First prize for a poster presented at the 2019 Canadian Mathematical Society winter meeting

2017 NSERC USRA
Undergraduate research on a Radon transform in Minkowski space, supervised by Dmitry Faifman

2013–2016 In-course scholarships
For academic excellence
- Isabel Bader In Course Scholarship
- William Ewart Staples Scholarship
- William Crichton Webster Scholarship
- University of Toronto Scholar

## Teaching

2025– **Instructor**, *Queen's University*, Kingston, Canada
In reverse chronological order
- MATH/MTHE 225: Ordinary Differential Equations
- APSC 172: Calculus II
- MATH/MTHE 225: Ordinary Differential Equations

2021–2022 **Instructor**, *University of Toronto*, Toronto, Canada
MAT235: Multivariable Calculus

2015–2023 **Teaching Assistant**, *University of Toronto*, Toronto, Canada
In reverse chronological order
- Hamiltonian Mechanics (graduate)
- Classical Geometries
- Differential Geometry
- Control Theory (graduate)
- Analysis I
- Calculus II
- Introduction to Proofs
- Introduction to Topology
- Introduction to Ordinary Differential Equations
- Calculus II
- Multivariable Calculus
- Calculus II
- Differential Topology (graduate)
- Calculus!
- Linear Algebra II

## Service

### Journal Referee or Reviewer

SIGMA: Symmetry, Integrability, and Geometry: Methods and Applications

Proceedings of the Royal Society of Edinburgh, Section A: Mathematics

Transactions of the American Mathematical Society

Israel Journal of Mathematics

Advances in Geometry

Pure and Applied Mathematics Quarterly

zbMATH Open

Mathematical Reviews

### Community

2025 Organizer of the workshop "Exloring new arrows in the BCWK groupoid II," a second-instalment of a retreat for early career researchers
March 7–10, 2025 in Burbach, Germany

2024 Organizer of the workshop "Exploring new arrows in the BGW groupoid," a retreat for early career researchers
October 25–28, 2024 in Bielefeld, Germany

2022–2023 Organizer of the Symplectic Seminar at the University of Toronto
Weekly

2022–2023 Organizer of the Graduate Student Seminar at the University of Toronto
Weekly

2021–2022 Mathematics representative to Graduate Student Union at the University of Toronto

2021–2022 Facilitator of a graduate student social form